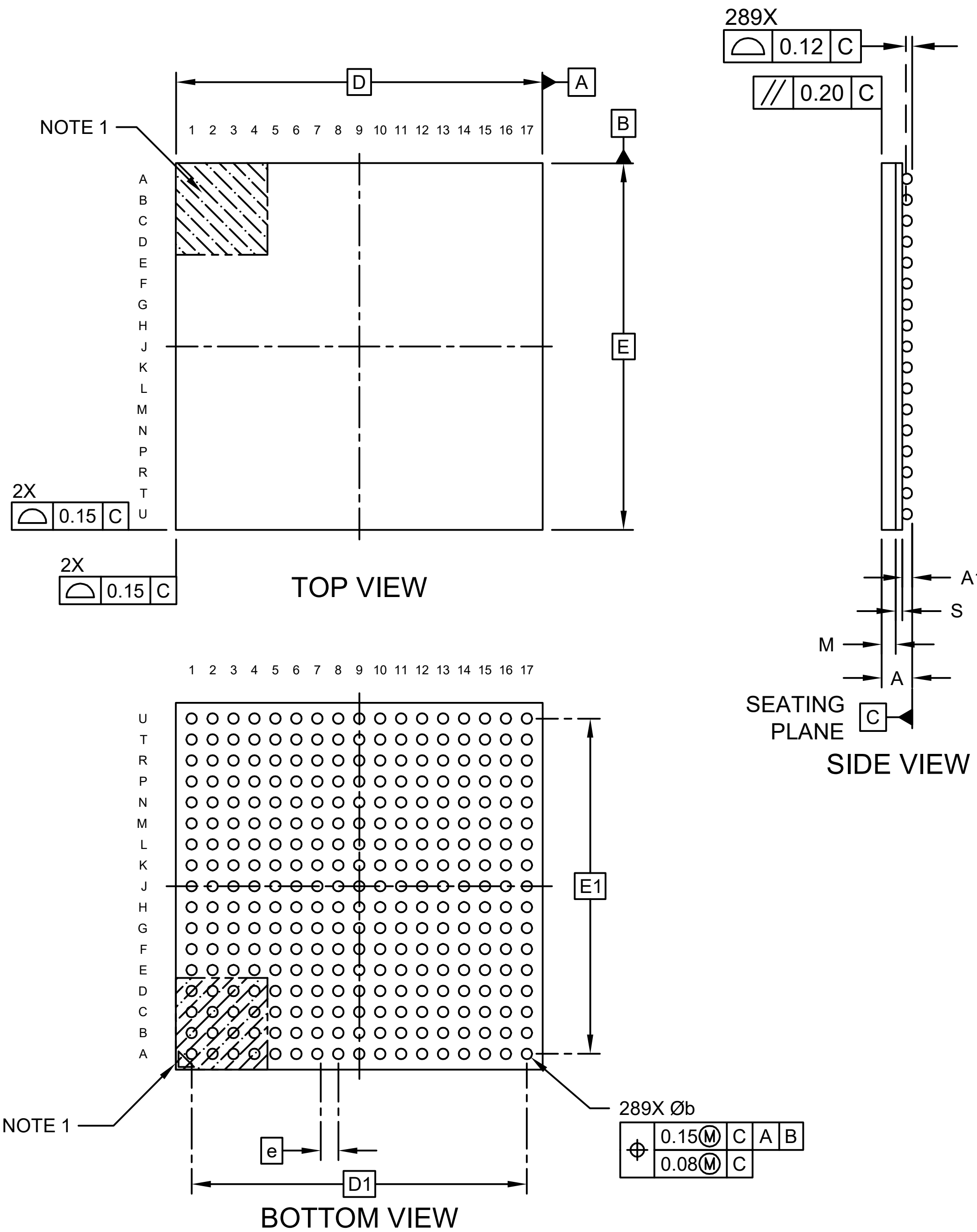
289-Ball Thin Fine-Pitch Ball Grid Array Package (4BB) - 14×14×1.2 mm Body [TFBGA]

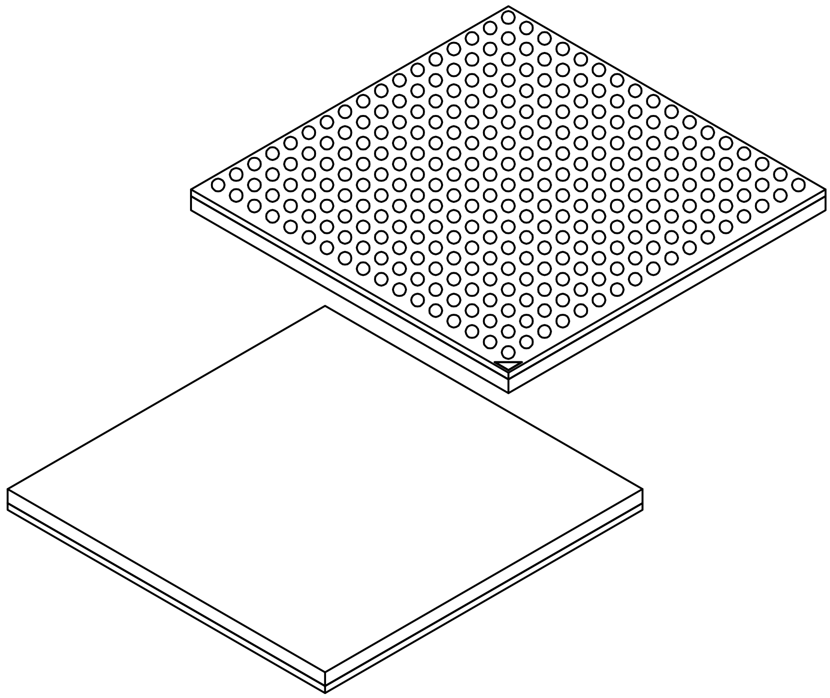
Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



Microchip Technology Drawing C04-21504 Rev A Sheet 1 of 2

289-Ball Thin Fine-Pitch Ball Grid Array Package (4BB) - 14×14×1.2 mm Body [TFBGA]

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



		Units	MILLIMETERS		
		Dimension Limits	MIN	NOM	MAX
Number of Terminals	N			289	
Pitch	e			0.80	
Overall Height	A		–	–	1.20
Ball Height	A1		0.27	–	0.37
Mold Cap Thickness	M			0.53 REF	
Substrate Thickness	S			0.26 REF	
Overall Length	D			14.00 BSC	
Ball Array Length	D2			12.80 BSC	
Overall Width	E			14.00 BSC	
Ball Array Width	E2			12.80 BSC	
Ball Width	b		0.38	–	0.48
Ball Diameter				0.40 REF	

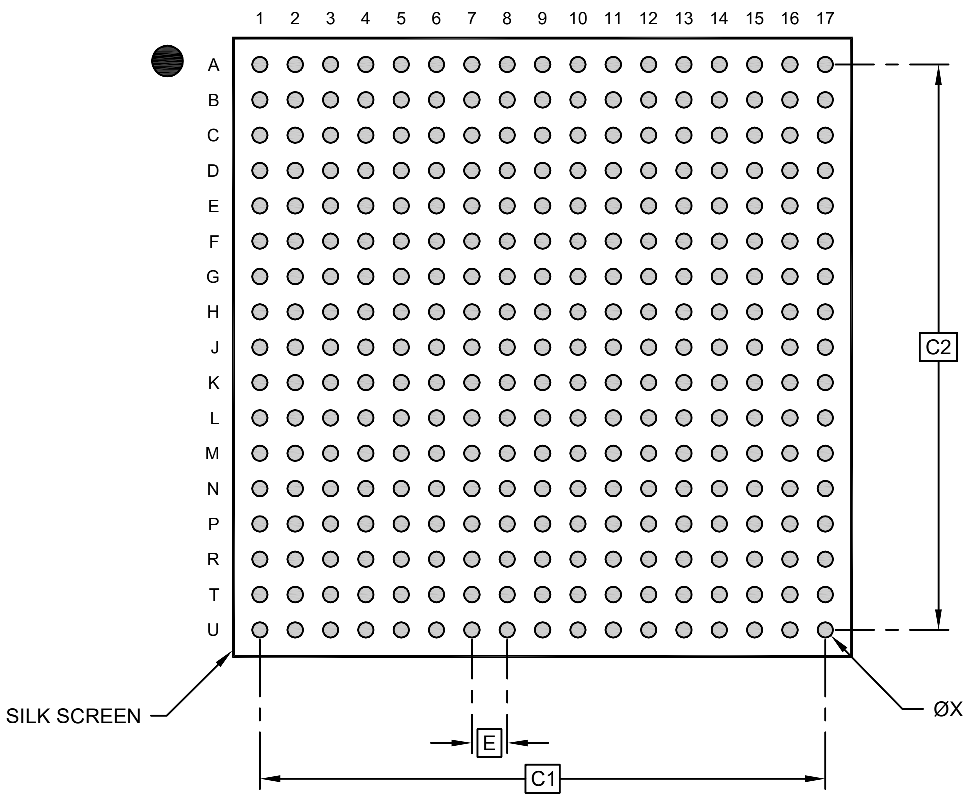
Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-21504 Rev A Sheet 2 of 2

289-Ball Thin Fine-Pitch Ball Grid Array Package (4BB) - 14×14×1.2 mm Body [TFBGA]

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

	Units	MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.80 BSC	
Contact Pad Spacing	C1		12.80 BSC	
Contact Pad Spacing	C2		12.80 BSC	
Contact Pad Diameter (X289)	X1			0.35

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

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